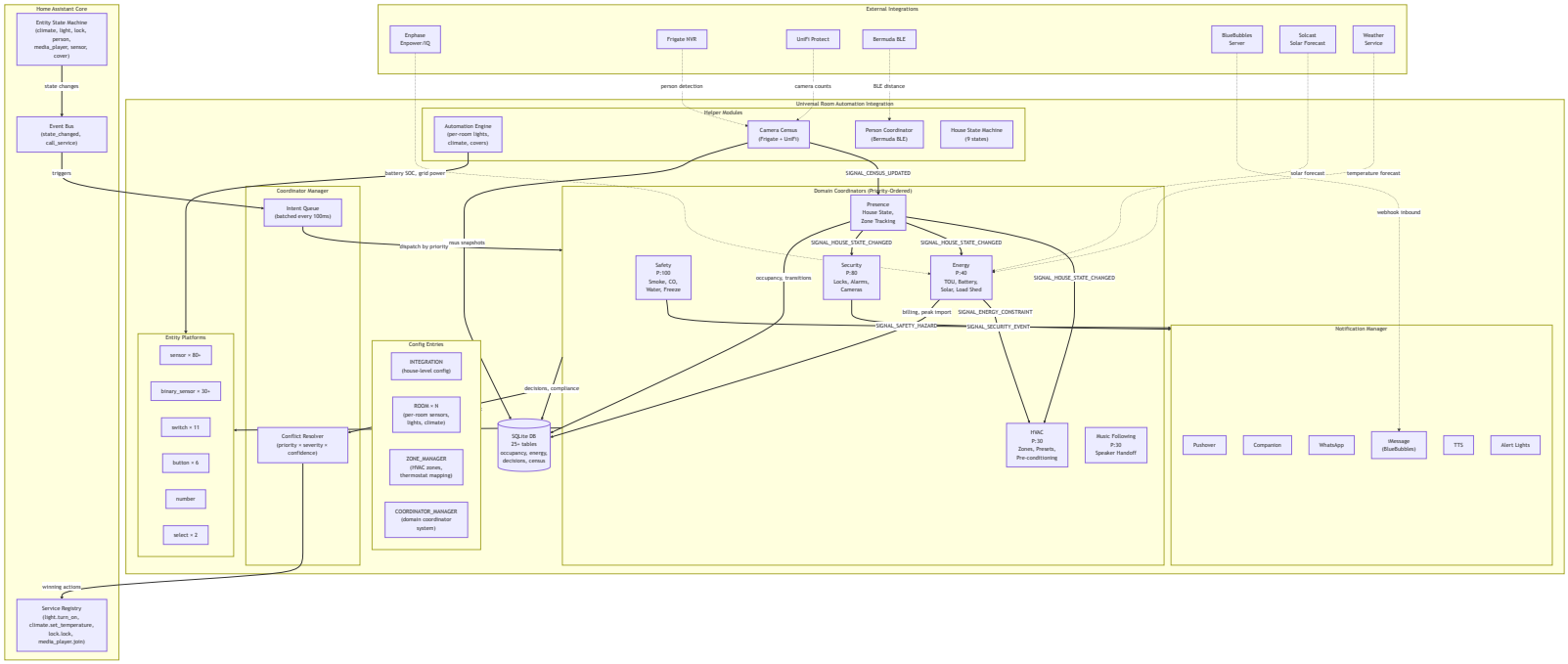
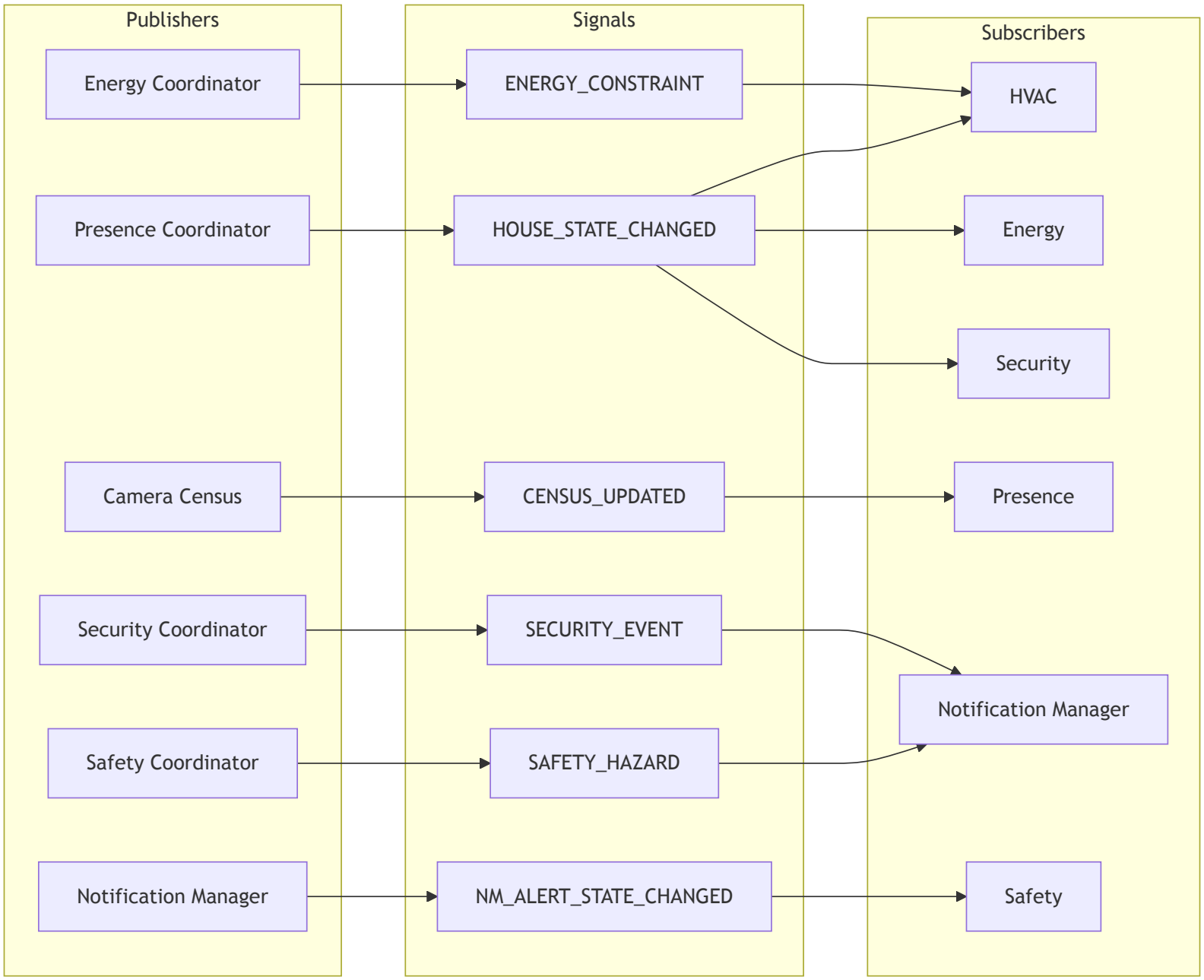
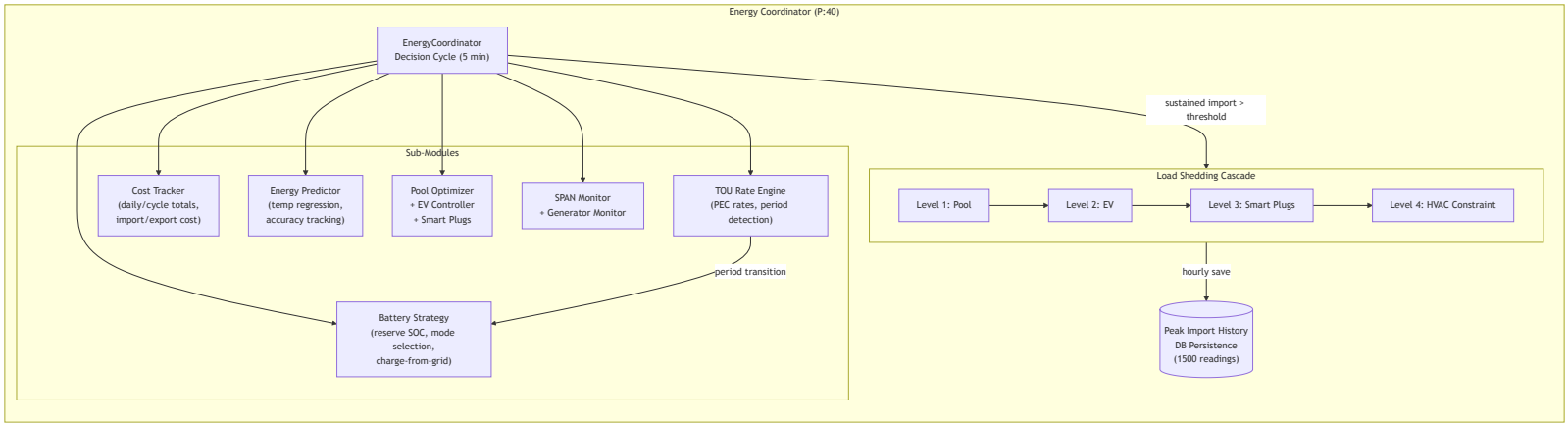
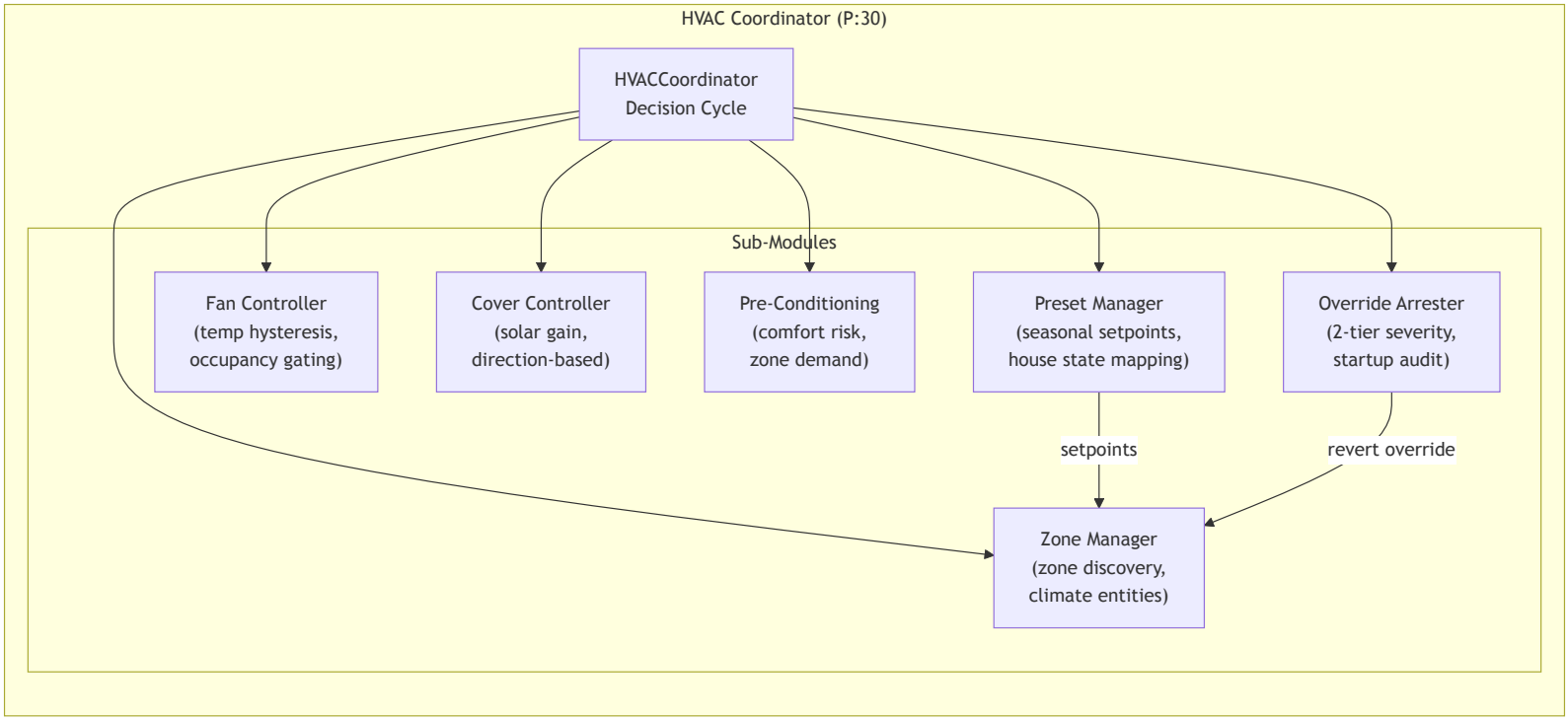
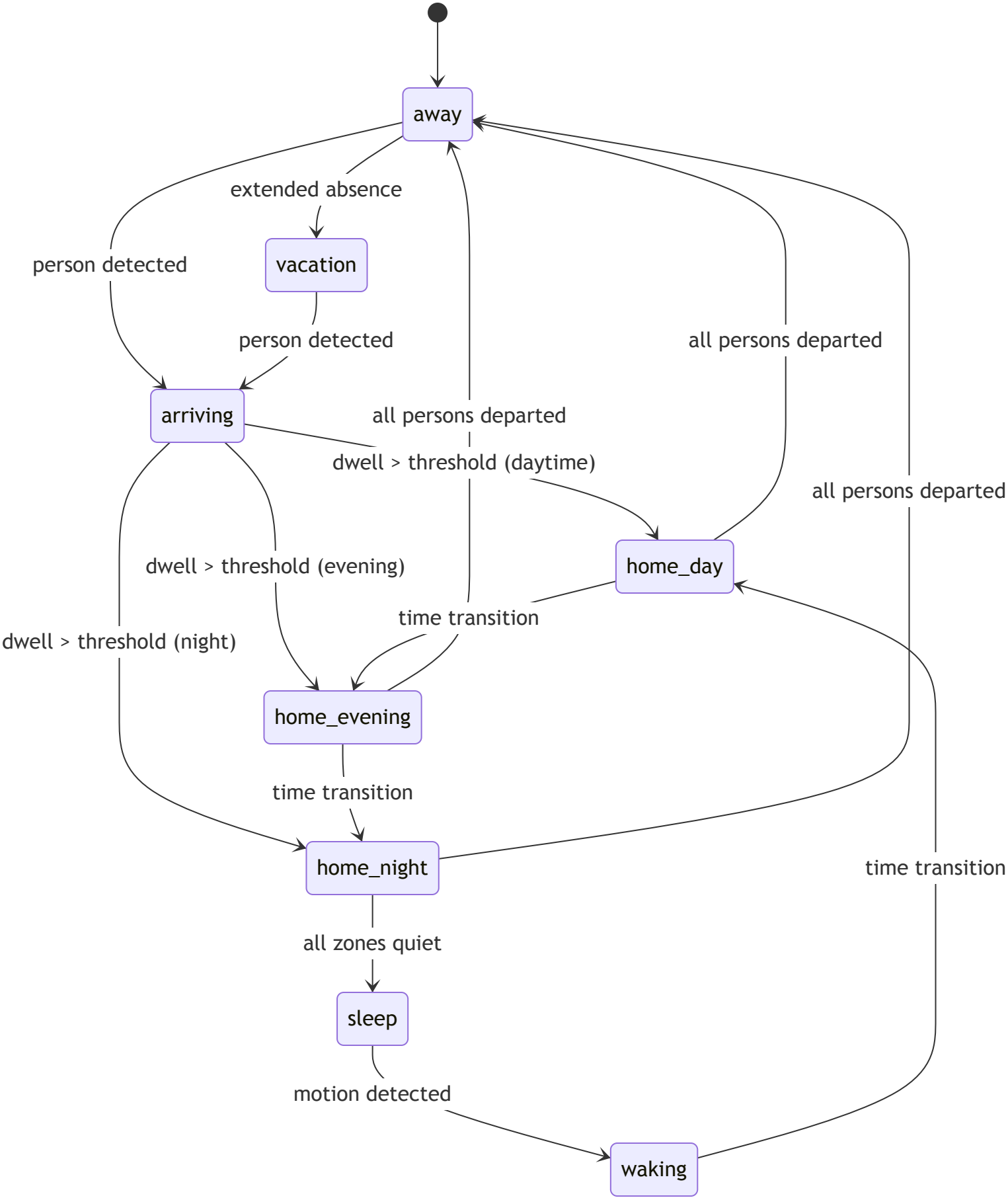
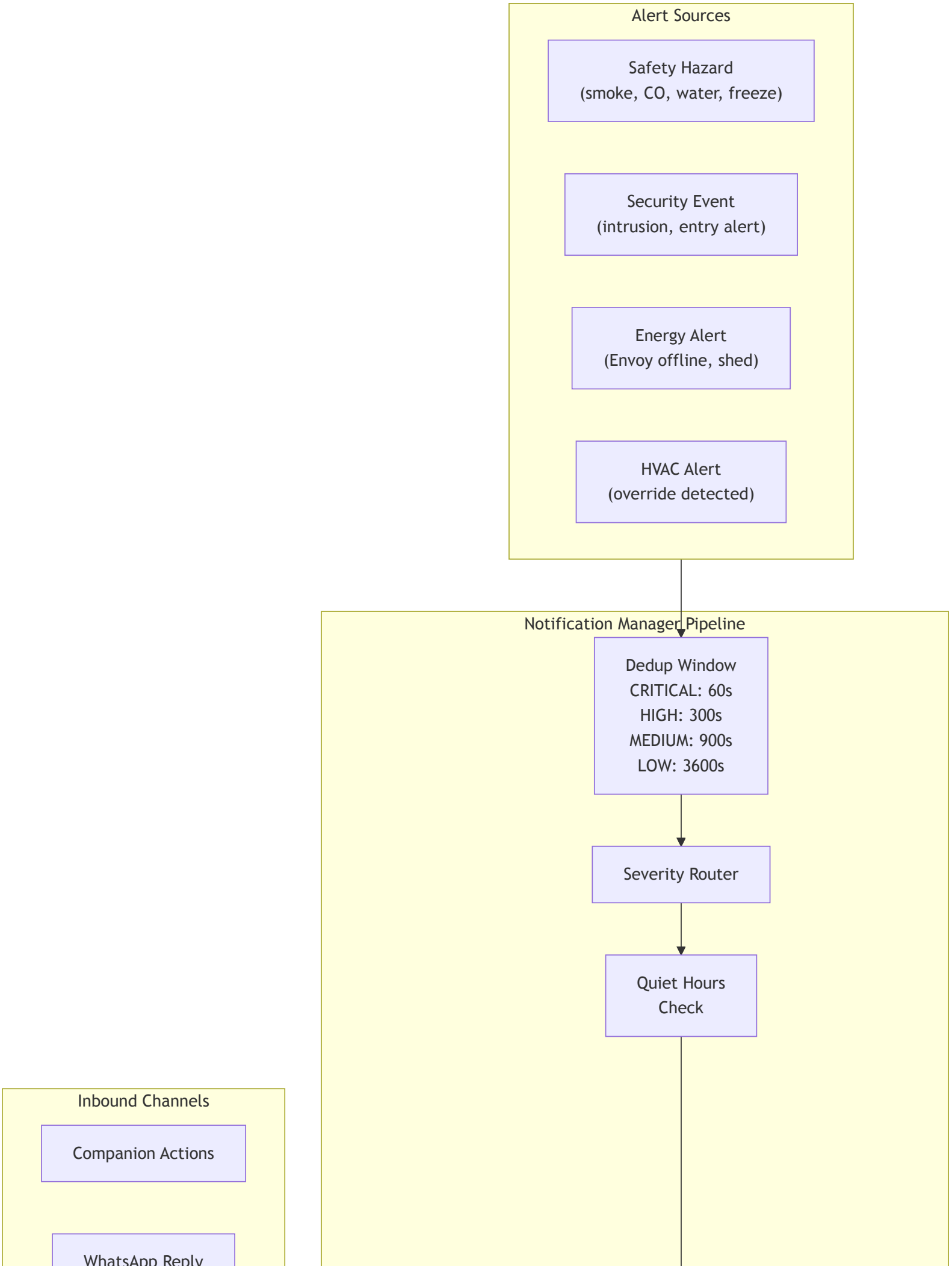

Home Assistant Core
Entity State Machine (climate, light, lock, person, media_player, sensor, cover)
state changes
Event Bus (state_changed, call_service)
triggers
Service Registry (light.turn_on, climate.set_temperature, lock.lock, media_player.join)
winning actions
External Integrations
Enphase Enpower/IQ
Frigate NVR
UniFi Protect
Bermuda BLE
BlueBubbles Server
Solcast Solar Forecast
Weather Service
person detection
camera counts
BLE distance
Universal Room Automation Integration
Helper Modules
Automation Engine (per-room lights, climate, covers)
Camera Census (Frigate + UniFi)
Person Coordinator (Bermuda BLE)
House State Machine (9 states)
battery SOC, grid power
SIGNAL_CENSUS_UPDATED
solar forecast
temperature forecast
webhook inbound
Coordinator Manager
Intent Queue (batched every 100ms)
dispatch by priority
room snapshots
Domain Coordinators (Priority-Ordered)
Presence House State, Zone Tracking
SIGNAL_HOUSE_STATE_CHANGED
SIGNAL_HOUSE_STATE_CHANGED
Safety P:100 Smoke, CO, Water, Freeze
Security P:80 Locks, Alarms, Cameras
Energy P:40 TOU, Battery, Solar, Load Shed
occupancy, transitions
SIGNAL_HOUSE_STATE_CHANGED
billing_peak import
SIGNAL_ENERGY_CONSTRAINT
SIGNAL_SAFETY_HAZARD
SIGNAL_SECURITY_EVENT
HVAC P:30 Zones, Presets, Pre-conditioning
Music Following P:30 Speaker Handoff
Entity Platforms
sensor × 80+
binary_sensor × 30+
switch × 11
button × 6
number
select × 2
Conflict Resolver (priority × severity × confidence)
Config Entries
INTEGRATION (house-level config)
ROOM × N (per-room sensors, lights, climate)
ZONE_MANAGER (HVAC zones, thermostat mapping)
COORDINATOR_MANAGER (domain coordinator system)
decisions, compliance
SQLite DB 25+ tables occupancy, energy, decisions, census
Notification Manager
Pushover
Companion
WhatsApp
iMessage (BlueBubbles)
TTS
Alert Lights

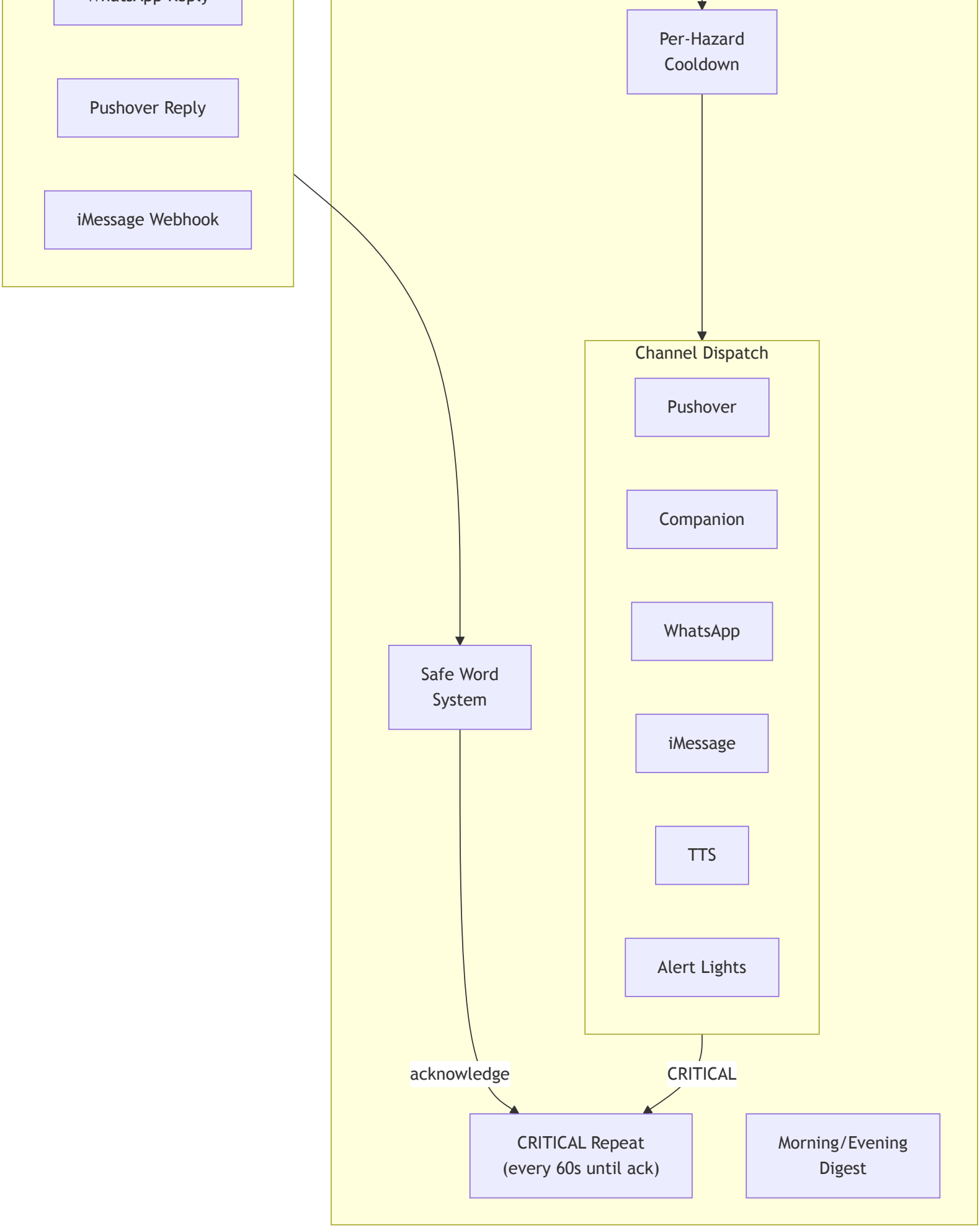

Publishers
Signals
Subscribers
Energy Coordinator
ENERGY_CONSTRAINT
HVAC
Presence Coordinator
HOUSE_STATE_CHANGED
Energy
Security
Camera Census
CENSUS_UPDATED
Presence
Security Coordinator
SECURITY_EVENT
Notification Manager
Safety Coordinator
SAFETY_HAZARD
Notification Manager
NM_ALERT_STATE_CHANGED
Safety


Energy Coordinator (P:40)
EnergyCoordinator
Decision Cycle (5 min)
sustained import > threshold
Sub-Modules
Cost Tracker
(daily/cycle totals,
import/export cost)
Energy Predictor
(temp regression,
accuracy tracking)
Pool Optimizer
+ EV Controller
+ Smart Plugs
SPAN Monitor
+ Generator Monitor
TOU Rate Engine
(PEC rates, period
detection)
period transition
Battery Strategy
(reserve SOC, mode
selection,
charge-from-grid)
Load Shedding Cascade
Level 1: Pool
Level 2: EV
Level 3: Smart Plugs
Level 4: HVAC Constraint
hourly save
Peak Import History
DB Persistence
(1500 readings)


HVAC Coordinator (P:30)
HVACCoordinator
Decision Cycle
Sub-Modules
Fan Controller
(temp hysteresis,
occupancy gating)
Cover Controller
(solar gain,
direction-based)
Pre-Conditioning
(comfort risk,
zone demand)
Preset Manager
(seasonal setpoints,
house state mapping)
Override Arrester
(2-tier severity,
startup audit)
setpoints
revert override
Zone Manager
(zone discovery,
climate entities)


away
extended absence
vacation
person detected
person detected
arriving
all persons departed
dwell > threshold (daytime)
all persons departed
all persons departed
home_day
dwell > threshold (evening)
time transition
dwell > threshold (night)
home_evening
time transition
home_night
time transition
all zones quiet
sleep
motion detected
waking


Alert Sources
Safety Hazard
(smoke, CO, water, freeze)
Security Event
(intrusion, entry alert)
Energy Alert
(Envoy offline, shed)
HVAC Alert
(override detected)
Notification Manager Pipeline
Dedup Window
CRITICAL: 60s
HIGH: 300s
MEDIUM: 900s
LOW: 3600s
Severity Router
Quiet Hours
Check
Inbound Channels
Companion Actions
WhatsApp Reply


Pushover Reply
iMessage Webhook
Per-Hazard Cooldown
Channel Dispatch
Pushover
Companion
WhatsApp
iMessage
TTS
Alert Lights
Safe Word System
acknowledge
CRITICAL
CRITICAL Repeat (every 60s until ack)
Morning/Evening Digest

OCCUPANCY_EVENTS		
string	room_id	PK
string	event_type	
float	timestamp	

PERSON_VISITS	
string	person_id
string	room_id
float	entry_time
float	exit_time
float	confidence

ROOM_TRANSITIONS	
string	person_id
string	from_room
string	to_room
float	timestamp
string	validation_method

ENERGY_DAILY		
string	date	PK
float	import_kwh	
float	export_kwh	
float	net_cost	
float	consumption_kwh	
float	predicted_consumption_kwh	
float	avg_temperature	

ENERGY_PEAK_IMPORT		
int	id	PK
int	seq	
float	import_kw	

DECISION_LOG	
float	timestamp
string	coordinator_id
string	situation
string	actions
string	scope

CENSUS_SNAPSHOTS	
float	timestamp
string	zone_id
int	frigate_count
int	unifi_count

NOTIFICATION_INBOUND		
int	id	PK
string	channel	
string	person_id	
string	message	
string	response_type	